

Introduction and Prerequisites

The following pages give high level level directions for setting up Intune with basic Mac management capabilities.

It is assumed that you are already familiar with Intune and have existing Intune and Apple Business Manager instances set up and have logins with sufficient privilege to carry out the operations required.

In Apple Business Manager a managed Apple ID with either the Administrator or Device Enrollment Manager role is required to add an MDM. To add a Location and assign the content Manager Role the Administrator or People Manager role is required.

In Intune sign in as a member of the *Global Administrator* or *Intune Service Administrator* Entra ID roles

Intune Setup

This section describes the essential setup information that links Intune to the various Apple Services needed to achieve Automated Device enrollment and ongoing device management

APNS Certificate

Apple Push Notification Services are critical to the operation of MDM. For any action the MDM needs to take such as issuing a command or installing a profile the initial communication with the device is via APNS.

To access the Intune UI, in a browser open <https://intune.microsoft.com/> and log in with a **Entra ID** user ID that has the role of *Global Administrator* or *Intune Service Administrator* for your tenancy

Navigate to **Devices -> macOS -> macOS Enrollment**

Click **Apple MDM Push Certificate**

Check the box granting permission to send information to Apple

Click on **Download your CSR**

In a new browser window or tab open <https://identity.apple.com/pushcert/> and log in with a managed Apple ID.

Do not use a personal Apple ID for this process as management the APNS certificate will be critical over the life of your device management solution and access via a personal Apple ID may become unavailable as staff change over time. Using a Managed Apple ID keeps control within the Apple Business Manager (or Apple School Manager) instance.

Click **Create a Certificate**

Upload the CSR, create and download the certificate

You can log out and close this window/tab

Return to the **Intune UI**

Under **Apple ID** Enter the Managed Apple ID you used to log in and create the certificate

Under **Apple MDM push certificate** Select the certificate you just downloaded then click **Upload**

The Microsoft Documentation for this procedure is at <https://learn.microsoft.com/en-us/mem/intune/enrollment/apple-mdm-push-certificate-get>

Automated Device Enrollment Token

The Device Enrollment token enables secure communication between Apple Business Manager or Apple School Manager and the MDM to allow for Automated Device Enrollment.

In the **Intune UI**

Navigate to **Devices -> iOS/iPadOS -> iOS/iPadOS Enrollment**

(Devices -> macOS -> macOS Enrollment will also work)

Click **Enrollment program tokens**

Click on **+ Add** to add a new token

Check the box granting permission to send information to Apple

Click on **Download your public key**

To access Apple Business Manager open a new browser window or tab and navigate to <https://business.apple.com> and log in with a Managed Apple ID that has the role of Administrator or Device Enrollment Manager

Click your name at the bottom of the sidebar, and click **Preferences**

Under **Your MDM Servers** click the plus sign to **+ Add**

Name your MDM server and upload the **Public Key**

Download the Server Token by clicking on **Download Token** then clicking **Download Server Token**

In the **Intune UI**

Under **Apple ID** enter the Managed Apple ID you used to log in and create the Server Token

Under **Apple token** select the Server Token just downloaded then click **Next** then click **Create**

The Microsoft Documentation for this procedure is at

<https://learn.microsoft.com/en-us/mem/intune/enrollment/device-enrollment-program-enroll-ios>

Apps and Books Token

The Apps and Books Token (previously known as the Volume Purchase Program or VPP token) enables secure communication between Apple Business Manager or Apple School Manager and the MDM for the management of Apps and Book licensing

In **Apple Business Manager**

In the sidebar click **Locations**

Click the plus sign to **+ Add**

Enter the details and create the new location

Click your name at the bottom of the sidebar, and click **Preferences**

Click **Payments and Billing**

Find the location you just created under **Server Tokens** and download the VPP token by clicking **Download**

In the **Intune UI**

Navigate to **Tenant administration -> Connectors and tokens -> Apple VPP Tokens**

Click **+ Create** to Create VPP Token

Under **Token Name** give the Token the same name used in ABM

Enter the Managed Apple ID used to create the VPP token in ABM

Upload the token file and click **Next**

Create VPP token ...

✓ Basics **2 Settings** ③ Scope tags ④ Review + create

Take control of token from another MDM ☒ No ☐ Yes



Country/Region:

Australia



Type of VPP account:

Business



Automatic app updates ⓘ

☐ No ☒ Yes



I grant Microsoft permission to send both user and device information to Apple. [Learn more](#)  *

Click through to complete the process.

You can ignore the Scope Tags page for this exercise

The Microsoft Documentation for this procedure is at

<https://learn.microsoft.com/en-us/mem/intune/apps/vpp-apps-ios>

iOS / iPadOS

iOS Device Enrollment Profile

The Device Enrollment Profile is created by the MDM and synchronised with Apple Business Manager or Apple School Manager to enable the Automated Device Enrolment process. A Device Enrollment Profile is created and assigned to one or more devices. Upon activation that profile will be installed on the device which will provide certificates and the URL of the MDM that will manage the device. A list of Setup Assistant items to hide from the user is also provided.

Purchase Company Portal licenses

In **Apple Business Manager**

In the sidebar click **Apps and Books**

Search for **Microsoft Intune Company Portal** and select it.

Under **Buy Licenses** select the location corresponding to your Intune instance and specify 2 in the quantity then click **Get**

NOTE: *Company Portal is Deployed automatically as part of AutomatedDevice Enrolment. It is a pre requisite that VPP licenses are available for this to work but it is currently not necessary to separately scope the app to the required devices. This will change in future.*

In the **Intune UI**

Ensure the app appears in the list under Apps -> iOS/iPadOS apps

Intune Company Portal

iOS volume purchase program app

NOTE: *It may be necessary to force synchronisation after the ABM app purchase*

In the **Intune UI**

Navigate to **Tenant administration -> Connectors and tokens -> Apple VPP Tokens**

Right click on the token name and select **Sync**

Navigate to **Devices -> iOS/iPadOS -> iOS/iPadOS Enrollment**

Click **Enrollment program tokens**

Click the token you created from the list

Under Manage click **Profiles**

Click **+ Create Profile -> iOS/iPadOS**

Complete the config as follows:

Name the profile e.g. iOS ADE with MFA then click **Next**

Settings should be as follows:

User affinity = Enroll with User Affinity

Authentication Method = Setup Assistant with modern authentication

Locked Enrollment = Yes

Await Final Configuration = Yes

Other settings can be default

Define enrollment and management settings for your iOS/iPadOS devices. [Learn more.](#)

User Affinity & Authentication Method

User affinity * ⓘ

Authentication Method ⓘ

 For devices running iOS/iPadOS 13.0 and later. For conditional access to work with this authentication method, Company Portal will be deployed to users as a required app to allow for device registration with Azure AD.

Install Company Portal with VPP ⓘ

Management Options

Supervised * ⓘ

 Supervision is required for devices using Company Portal as their authorization method.

Locked enrollment * ⓘ

Sync with computers: * ⓘ

Apple Configurator certificates: ⓘ 

Uploaded Certificates

No certificates, select a certificate file to import.

Await final configuration ⓘ ☒ Yes ☐ No

Device Name

Apply device name template (supervised only) ⓘ ☐ Yes ☒ No

Cellular data plan

Activate cellular data ⓘ ☐ Yes ☒ No

Enter a department Name and Phone Number and toggle all setup assistant screens to Hide except

Passcode = Show

Location Services = Show

Touch ID and Face ID = Show

1 Setup Assistant

2 Review + save

Department * ⓘ

Contoso IT



Department Phone * ⓘ

1 800 123456



Setup Assistant Screens ⓘ

Toggle All

Passcode

Hide

Show

Location Services

Hide

Show

Restore

Hide

Show

Apple ID

Hide

Show

Terms and conditions

Hide

Show

Touch ID and Face ID

Hide

Show

Apple Pay

Hide

Show

Zoom

Hide

Show

Siri

Hide

Show

Diagnostics Data

Hide

Show

Display Tone

Hide

Show

Privacy

Hide

Show

Android Migration

Hide

Show

Home Button

Hide

Show

iMessage & FaceTime

Hide

Show

Onboarding

Hide

Show

Screen Time

Hide

Show

SIM Setup

Hide

Show

Software Update

Hide

Show

Watch Migration

Hide

Show

Appearance

Hide

Show

Device To Device Migration	<input checked="" type="button" value="Hide"/> Show
Restore Completed	<input checked="" type="button" value="Hide"/> Show
Software Update Completed	<input checked="" type="button" value="Hide"/> Show
Get Started	<input checked="" type="button" value="Hide"/> Show
Terms of Address	<input checked="" type="button" value="Hide"/> Show

Save the enrolment profile

Click **Set default profile** and set the just created profile as the default for iOS/iPadOS

iOS Configuration Profiles

Create a few essential configuration profiles. The method is the same in each case with one possible key difference. Use of a Template or the Setting catalogue.

Passcode Compliance Policy

Setting a Compliance Policy creates a configuration profile and also implements a corresponding compliance check in Intune. This policy is used to ensure a device passcode is set with appropriate complexity.

<https://learn.microsoft.com/en-us/mem/intune/protect/compliance-policy-create-ios>

In the **Intune UI**

Navigate to **Devices -> iOS/iPadOS -> Compliance Policies**

Click **+ Create** to Create a new policy then Create a Policy **Create**

Give the policy a name e.g. Passcode then click **Next**

Under **System Security**

Password

Require a password to unlock mobile devices ⓘ

Require

Not configured

Device enrollment and automated device enrollment

These settings work for devices that were enrolled in Intune through device enrollment, and for devices enrolled using Apple School Manager or Apple Business Manager with automated device enrollment (formerly DEP).

Simple passwords ⓘ

Block

Not configured

Minimum password length ⓘ

6



Required password type ⓘ

Numeric



Number of non-alphanumeric characters in password ⓘ

Not configured



Maximum minutes after screen lock before password is required ⓘ

Not configured



Maximum minutes of inactivity until screen locks ⓘ

Not configured



Password expiration (days) ⓘ

Enter number of days (1-730)

Number of previous passwords to prevent reuse ⓘ

Enter a number (1-24)

Device Security

Restricted apps ⓘ

Export

App name

App bundle ID

Click **+ Add all devices**

Click **Next**

Click **Create**

Managed Open In Profile

The Managed Open profile is used to implement the Managed Open In controls on iOS and iPadOS that limit the flow of data between managed and unmanaged apps.

In the **Intune UI**

Navigate to **Devices -> iOS/iPadOS -> Configuration Profiles**

Click **+ Create profile** to Create a new profile

Click **Profile Type - Settings Catalogue** then **Create**

Give the profile a name e.g. Managed Open In then click **Next**

Click **+ Add settings** and choose **Restrictions** and select the settings in the screenshot below and close the **Settings Picker**

Set the values to True or False as shown

^ Restrictions Remove category

Configure the Restrictions payload to enable or disable features on devices. These configurations can be used prevent users from accessing a specific app, service or function on enrolled devices. For example, a restriction can be added that prevents an iPhone or iPad from using AirPrint. Another restriction can be added to prevent the sharing of passwords over AirDrop on an iPhone, iPad and Mac. Certain restrictions on an iPhone may be mirrored on a paired Apple Watch.

 154 of 158 settings in this category are not configured

Require Managed Pasteboard ⓘ	☒ True	⊖
Allow Unmanaged To Read Managed Contacts ⓘ	☒ True	⊖
Allow Open From Unmanaged To Managed ⓘ	☐ False	⊖
Allow Open From Managed To Unmanaged ⓘ	☐ False	⊖

Click **Next** then on Scope Tags click **Next**

In Assignments Click **+ Add all devices**

Click **Next** then Click **Create**

Single Sign On Profile

The Single Sign On profile enables the use of Apple's Extensible Single Sign On framework in conjunction with the Entra ID SSO Extension to reduce the burden on users logging in while maintained strong authentication and authorisation using modern authentication protocols. This also requires that Microsoft Authenticator is installed on the device.

*In the **Intune UI**

Navigate to **Devices -> iOS/iPadOS -> Configuration Profiles**

Click **+ Create profile** to Create a new profile

Click **Profile Type - Settings Catalogue** then **Create**

Give the profile a name e.g. Entra ID SSO then click Next

Click **+ Add settings** and choose **Authentication -> Extensible Single Sign On (SSO)** and select the settings in the screenshot below and close the Settings Picker

Extensible Single Sign On (SSO)

[Remove subcategory](#)

Configure an app extension that enables single sign-on (SSO) for devices.

 3 of 13 settings in this subcategory are not configured

URLs

 Delete  Sort  Import  Export

☐

https://login.microsoftonline.com

✓

☐

https://login.microsoft.com

✓

☐

https://sts.windows.net

✓

☐

https://login.partner.microsoftonline.cn

✓

☐

https://login.chinacloudapi.cn

✓

☐

https://login.microsoftonline.us

✓

☐

https://login-us.microsoftonline.com

✓

Extension Data

 Add  Delete

☐ Key 

Configure settings

☐ browser_sso_interaction_enabled

[+ Edit instance](#)

☐ disable_explicit_app_prompt

[+ Edit instance](#)

☐ AppPrefixAllowList

[+ Edit instance](#)

☐ device_registration

[+ Edit instance](#)

Extension Identifier 

com.microsoft.azureauthenticator.ssoextension ✓

Type * 

Redirect ▾

Key	Type	Value
browser_sso_interaction_enabled	Integer	1

Key	Type	Value
disable_explicit_app_prompt	Integer	1
AppPrefixAllowList	String	com.microsoft.,com.apple.
device_registration	String	{{DEVICEREGISTRATION}}

Click **Next** then on Scope Tags click **Next**

In Assignments Click + **Add all devices**

Click **Next** then Click **Create**

Email Profile

The Email profile will add an account the the native iOS Mail app on behalf of the user.

In the **Intune UI**

Navigate to **Devices -> iOS/iPadOS -> Configuration Profiles**

Click + **Create profile** to Create a new profile

Click **Profile Type - Templates** then **Email** then **Create**

Give the profile a name e.g. Exchange Online then click **Next**

Exchange ActiveSync account settings

Email server * ⓘ

m.outlook.com✓

Account name * ⓘ

Exchange Online✓

Username attribute from AAD * ⓘ

User principal name▼

Email address attribute from AAD * ⓘ

Primary SMTP Address▼

Authentication method * ⓘ

Username and password▼

SSL ⓘ

EnableDisable

OAuth ⓘ

EnableDisable

Exchange ActiveSync profile configuration

Exchange data to sync ⓘ

All data▼

Allow users to change sync settings ⓘ

Yes▼

Exchange ActiveSync email settings

S/MIME ⓘ

Disable S/MIME▼

Amount of email to synchronize ⓘ

Not configured▼

Allow messages to be moved to other email accounts ⓘ

EnableDisable

Allow email to be sent from third party applications ⓘ

EnableDisable

Synchronize recently used email addresses ⓘ

EnableDisable

VPN profile for per account VPN ⓘ

None▼

Click **+** **Add all devices**

Click **Next**

Click **Create**

iOS App Deployment

VPP

Deploy Microsoft Authenticator

In **Apple Business Manager**

In the sidebar click **Apps and Books**

Search for **Microsoft Authenticator** and select the iOS version

Under Buy Licenses select the location corresponding to your Intune instance and specify 10 in the quantity then click **Get**

In the Intune UI

Navigate to **Apps -> iOS/iPadOS**

Select **Microsoft Authenticator** from the list of available apps. If it is not listed try clicking Refresh

Click on **Properties**

Click on Assignments **Edit** and assign the app under Required -> **+ Add all devices** then **Review + save**

Deploy Additional Apps

Repeat the above process to deploy additional apps such as Microsoft Teams and OneDrive and Workday

You can make apps appear as optional installs within the Company Portal app by assigning them as Available for enrolled devices rather than Required

NOTE: It may be necessary to force synchronisation after the ABM app purchase

In the Intune UI

Navigate to **Tenant administration -> Connectors and tokens -> Apple VPP Tokens**

Right click on the token name and select **Sync**

App Configuration

These settings are app specific and control, behaviour of the specific apps as determined by the app developer. The example below sets the UPN used by Teams based on the user that enrolled the device

Teams

In the Intune UI

Apps -> App Configuration Policies

Click **+ Add -> Managed Devices**

Give the policy a name e.g. Teams UPN

Select Platform **iOS/iPadOS**

Targeted App -> Select **Microsoft Teams**

Click **Next**

Configuration settings format -> select **Use configuration designer**

Enter values as follows:

Key	Type	Value
IntuneMAMAllowedAccountsOnly	String	Enabled
IntuneMAMUPN	String	{{userprincipalname}}

Create app configuration policy ...

1 Basics 2 Settings 3 Assignments 4 Review + create

Name * Teams ✓

Description

Device enrollment type Managed devices ▼

Platform * ⓘ iOS/iPadOS ▼

Targeted app * ⓘ Microsoft Teams

Set Scope to All devices

Click **Next**

Click **Create**

iOS Device Assignment

In **Apple Business Manager**

From the sidebar click Devices

Search for the serial number of the iPhone you will enrol and select it

Click **Edit MDM Server**

Click **Assign the Following MDM** and choose the appropriate MDM from the drop down list then click Continue

In the Intune UI

Navigate to **Devices -> iOS/iPadOS -> iOS/iPadOS Enrollment**

Click **Enrollment program tokens**

Select the token you created from the list

Under Manage click **Devices**

If the device you assigned from ABM is not listed click Sync, wait a minute and click Refresh

Check the box next to the device(s) in the list that you wish to apply an ADE profile then click **Assign Profile**

Under Assign Profile select the appropriate enrolment profile from the drop down list

Click **Sync** to sync with ABM. This can be done only after 15 mins or more from the last manual sync.

Enrolment Test

You can now test the enrolment process on the iPhone you have assigned

macOS

Mac Device Enrollment Profile

The Device Enrollment Profile is created by the MDM and synchronised with Apple Business Manager or Apple School Manager to enable the Automated Device Enrolment process. A Device Enrollment Profile is created and assigned to one or more devices. Upon activation that profile will be installed on the device which will provide certificates and the URL of the MDM that will manage the device. A list of Setup Assistant items to hide from the user is also provided.

In the **Intune UI**

Navigate to **Devices -> macOS -> macOS Enrollment**

Click **Enrollment program tokens**

Click the token you created from the list

Under Manage click **Profiles**

Click **+ Create Profile -> macOS**

Complete the config as follows:

Name the profile e.g. macOS ADE with MFA

User affinity = Enroll with User Affinity

Authentication Method = Setup Assistant with modern authentication

Locked Enrollment = Yes

Enter a department Name and Phone Number and toggle all setup assistant screens to Hide except

Location Services = Show

Touch ID and Face ID = Show

Configure Account Settings

1 Account Settings

2 Review + save

i If you create a local account, Await final configuration is set to Yes regardless of any other configuration even if the toggle for this setting in Management settings displays No. [Learn more about local account management](#)

Local primary account (preview)

Create a local primary account *	Yes
Prefill account info ⓘ	☒ Yes ☐ Not configured
Primary account name * ⓘ	{{partialupn}}
Supported variables: {{partialupn}}	
Primary account full name * ⓘ	{{username}}
Supported variables: {{username}}	
Restrict editing ⓘ	☒ Yes ☐ Not configured

Click **Create**

Click **Set default profile** and set the just created profile as the default for macOS

Set default enrollment profile

Select the enrollment profile to set as default profile. Set a default profile for iOS/iPadOS and macOS if you will be enrolling those platforms.

iOS/iPadOS Enrollment Profile

macOS Enrollment Profile

OK

Cancel

Mac Configuration Profiles

Create a few essential configuration profiles. The method is the same in each case with one possible key difference. Use of a Template of the Setting catalogue. An example of each is given.

For configuration profiles created using the Settings Catalogue there is an Export JSON capability. There is a corresponding capability to select Import Policy as an option when creating a new configuration profile so previous configs can be reused.

FileVault Profile

This profile enables FileVault full disk encryption.

In the **Intune UI**

Navigate to **Devices -> macOS -> Configuration Profiles**

Click **+ Create** to Create a new profile

Click **Profile Type - Settings Catalogue** then **Create**

Give the profile a name e.g. FileVault then click **Next**

Click **+ Add settings** and choose **Full Disk Encryption** and select the settings in the screenshot below and close the Settings Picker.

Make the selections as shown

FileVault

Configure the FileVault payload to manage FileVault disk encryption settings on devices.

i 6 of 12 settings in this subcategory are not configured

Enable * i	On	
Force Enable In Setup Assistant i	☒ True	⊖
Show Recovery Key i	☒ Enabled	⊖
Use Keychain i	☐ Disabled	⊖
Use Recovery Key i	Enabled	⊖
User Enters Missing Info i	☐ False	⊖

FileVault Options

Configure the FileVault Options payload to customize FileVault disk encryption settings on devices.

Prevent FileVault From Being Disabled i	☒ True	⊖
Prevent FileVault From Being Enabled i	☐ False	⊖

FileVault Recovery Key Escrow

Configure the FileVault Recovery Key Escrow payload to customize FileVault Recovery Key Escrow settings on devices.

i 1 of 2 settings in this subcategory are not configured

Location i	Contoso IT
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Click **Next** then on Scope Tags click **Next**

In Assignments Click **+ Add all devices**

Click **Next** then Click **Create**

Firewall Profile

This profile enables the built in macOS Application Firewall

In the **Intune UI**

Navigate to **Devices -> macOS -> Configuration Profiles**

Click + **Create** to Create a new profile

Click **Profile Type - Settings Catalogue** then **Create**

Give the profile a name e.g. Firewall then click **Next**

Click + **Add settings** and choose **Networking -> Firewall** and select the settings in the screenshot below and close the **Settings Picker**

Make the selections as shown

^ Networking

Remove category

Firewall

Remove subcategory

Configure the Firewall payload to turn on the Firewall in macOS to prevent unwanted connections from the internet or other networks.

 11 of 14 settings in this subcategory are not configured

Block All Incoming ⓘ

True

⊖

Enable Firewall ⓘ

True

Enable Stealth Mode ⓘ

True

⊖

Click **Next** then on Scope Tags click **Next**

In Assignments Click + **Add all devices**

Click **Next** then **** Click **Create**

Gatekeeper Profile

*This profile enables Gatekeeper to ensure only trusted apps can launch**on macOS*

In the **Intune UI**

Navigate to **Devices -> macOS -> Configuration Profiles**

Click **+ Create** to Create a new profile

Click **Profile Type - Settings Catalogue** then **Create**

Give the profile a name e.g. Gatekeeper then click **Next**

Click **+ Add settings** and choose setting from **App Store, System Policy** and select the settings in the screenshot below and close the **Settings Picker**

App Store

[Remove category](#)

Use the App Store payload to configure macOS App Store restrictions on devices.

i 2 of 3 settings in this category are not configured

Restrict Store Software Update Only ⓘ ☒ True



System Policy

[Remove category](#)

System Policy Managed

[Remove subcategory](#)

Configure the System Policy Managed payload to configure the Finder's contextual menu to bypass the system policy.

Disable Override ⓘ ☒ True



System Policy Control

[Remove subcategory](#)

i 2 of 2 settings in this subcategory are not configured

Enable Assessment ⓘ ☒ True



Click **Next** then on Scope Tags click **Next**

In Assignments Click + **Add all devices**

Click **Next** then Click **Create**

Extensible SSO Profile

The Single Sign On profile enables the use of Apple's Extensible Single Sign On framework in conjunction with the Entra ID SSO Extension to reduce the burden on users logging in while maintained strong authentication and authorisation using modern authentication protocols. This also requires that Company Portal is installed on the device.

In the **Intune UI**

Navigate to **Devices -> macOS -> Configuration Profiles**

Click **+ Create** to Create a new profile

Click **Profile Type - Settings Catalogue** then **Create**

Give the profile a name e.g. Entra ID SSO then click **Next**

Click **+ Add settings** and choose **Authentication -> Extensible Single Sign On (SSO)** and select the settings in the screenshot below and close the **Settings Picker**

Extensible Single Sign On (SSO)

Configure an app extension that enables single sign-on (SSO) for devices.

29 of 35 settings in this subcategory are not configured

Extension Data

+ Add Delete

☐ Key	Configure settings	
☐ browser_sso_interaction_enabled	+ Edit instance	
☐ disable_explicit_app_prompt	+ Edit instance	
☐ AppPrefixAllowList	+ Edit instance	

Extension Identifier com.microsoft.CompanyPortalMac.ssoextension

Team Identifier UBF8T346G9

Type * Redirect

URLs

Delete Sort + Import Export

☐	https://login.microsoft.com	
☐	https://sts.windows.net	
☐	https://login.partner.microsoftonline.cn	
☐	https://login.chinacloudapi.cn	
☐	https://login.microsoftonline.us	
☐	https://login-us.microsoftonline.com	

Key	Type	Value
browser_sso_interaction_enabled	Integer	1

Key	Type	Value
disable_explicit_app_prompt	Integer	1
AppPrefixAllowList	String	com.microsoft.,com.apple.

Click **Next** then on Scope Tags click **Next**

In Assignments Click **+ Add all devices**

Click **Next** then Click **Create**

Office 365 Profile

*This sets some parameters for Office 365 including the default user for**licensing*

In the **Intune UI**

Navigate to **Devices -> macOS -> Configuration Profiles**

Click **+ Create** to Create a new profile

Click **Profile Type - Settings Catalogue** then **Create**

Give the profile a name e.g. Office 365 Parameters then click **Next**

Click **+ Add settings** and choose **Microsoft Office -> Microsoft Office** then click the **Enable Automatic sign-in** and **Office Activation Email Address** checkboxes and close the **Settings Picker**

Microsoft Office

[Remove subcategory](#)

Configure the Microsoft Office preference settings to control diagnostic data that is collected and sent to Microsoft about Office client software being used as well as connected experiences that use cloud-based functionality to provide enhanced Office features to you and your users.

 18 of 20 settings in this subcategory are not configured

Enable automatic sign-in 

☒ True



Office Activation Email Address 



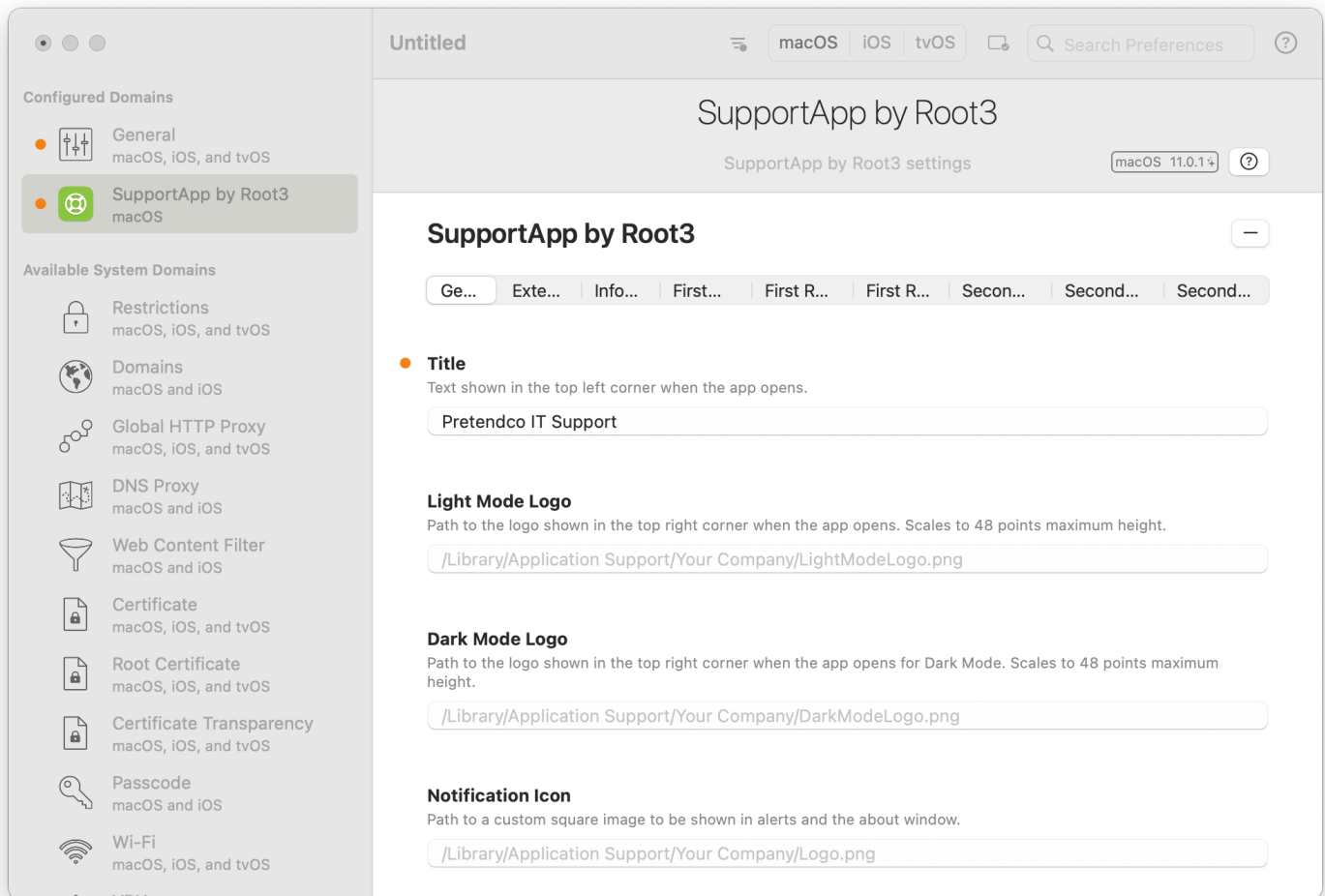
Under **Assignments** click **+Add all devices**

Custom Profile

This profile sets the configuration of the Support App which we will install as part of this setup.

Create a custom config for the Support app as per the documentation on the developers web site

This can be done with a text editor or a tool like [iMazing Profile editor](#).



In the **Intune UI**

Navigate to **Devices -> macOS -> Configuration Profiles**

Click + **Create** to Create a new profile

Click **Profile Type -> Templates** then **Custom** then **Create**

Give the profile a name e.g. Support App Config then click **Next**

For **Custom configuration profile name** enter Support App Config

Set **Deployment Channel** to **Device Channel**

Upload the Custom Configuration profile file. Use filename `Support_Intune.mobileconfig`

Under **Assignments** click **+Add all devices**

Managed Login Items Profile

Any background items that run upon a device boot or at user login including Launch Daemons, Launch Agents and Open at Login Apps can be deactivated by an Admin user. In order to prevent these items being deactivated a Managed Login Items config profile is deployed.

In the **Intune UI**

Navigate to **Devices -> macOS -> Configuration Profiles**

Click **+ Create** to Create a new profile

Click **Profile Type - Settings Catalogue** then **Create**

Give the profile a name e.g. Background and Login Items then click **Next**

Click **+ Add settings** and choose **Login -> Service Management - Managed Login Items** then check all but **Team Identifier** and close the **Settings Picker**

Setting name		
✓	☒ Rules	i
✓	☒ Rules	i
	☒ Comment	i
	☒ Rule Type	i
	☒ Rule Value	i
	☐ Team Identifier	i

Enter the following values:

Comment	Rule Type	Rule Value	Team Identifier
Microsoft Apps	Bundle Identifier Prefix	com.microsoft	Not Configured
Microsoft Daemons	Label Prefix	com.microsoft	Not Configured
Support App	Label	nl.root3.support	Not Configured

Under **Assignments** click **+Add all devices**

Mac App Deployment

Scripting

Deploy Company Portal

Company Portal is required for a number of functions. These include device registration with Azure AD, a self Service App catalogue and provision of the Azure AD SSO Extension code.

In a new browser window or tab open <https://github.com/microsoft/shell-intune-samples/blob/master/macOS/Apps/Company%20Portal/installCompanyPortal.zsh>

Copy the contents of the zsh script, paste in to a text editor and save the file as `InstallCompanyPortal.zsh`

In the **Intune UI**

Navigate to **Devices -> macOS -> Shell Scripts**

Click **+ Add**

Name the Script e.g. Install Company Portal and click **Next**

Upload the saved `InstallCompanyPortal.zsh` file and set the parameters as

Run script as signed-in user ⓘ

Yes No

Hide script notifications on devices ⓘ

Not configured

Script frequency ⓘ

Not configured

Max number of times to retry if script fails

3 times

Under **Assignments** click **+Add all devices**

Line of Business / Packaged

Deploy Microsoft Office

In the **Intune UI**

Navigate to **Apps -> macOS**

Click **+Add**

From the **App type** drop down list select **Microsoft 365 Apps / macOS**

Under **Assignments** click **+Add all devices** under **REQUIRED**

Deploy Support App

The Support App helps an the end user get quick access to basic device information and apps needed when receiving support from IT.

This package has been created and signed by the developer so can be uploaded and deployed by Intune.

Intune supports two methods to deploy .pkg files. In this example we will use the Line-of-business method that leverages the MDM command InstallEnterpriseApplication.

Download the latest packaged version of the Support app from

<https://github.com/root3nl/SupportApp/releases>

In the **Intune UI**

Apps -> macOS

Click **+ Add**

From the drop down menu select **Other -> Line-of-business app**

Upload the Support App package file `Support.x.x.pkg`

Select file * ⓘ

Support.2.5.pkg

Name * ⓘ

Support

Description * ⓘ

Support App

Publisher * ⓘ

Enter a publisher name

i The minimum operating system for uploading a .pkg file is macOS 10.14. Upload a .pkg file to select an older minimum operating system.

Minimum operating system * ⓘ

macOS Monterey 12.0

Ignore app version ⓘ

Yes

No

Install as managed ⓘ

Yes

No

Included apps * ⓘ

i Review the included apps list to edit apps or remove anything that isn't an app. The app listed first is used as the primary app in app reporting. The app version can be the CFBundleShortVersionString or CFBundleVersion. [Learn more about included apps.](#)

App bundle ID (CFBundleIdentifier)	App version (CFBundleShortVersionString)	
nl.root3.support	2.5	

Category ⓘ

Computer management

Show this as a featured app in the Company Portal ⓘ

Yes

No

Information URL ⓘ

Enter a valid url

Privacy URL ⓘ

Enter a valid url

Developer ⓘ

Owner ⓘ

Notes ⓘ

Logo ⓘ

Select image

Under **Assignments** click **+Add all devices**

Deploy Privileges App

The SAP privileges App helps an Admin user manage their risk by allowing them to demote themselves to a Standard user when having Admin privileges is not required

Download the latest version of the SAP Privileges tool from <https://github.com/SAP/macOS-enterprise-privileges>

On a Mac install Privileges in the Applications folder

Use the [Packages App](#) to create a file called Privileges.pkg

In the **Intune UI**

Apps -> macOS

Click **+Add**

From the drop down menu select **macOS app (PKG)**

Upload the .pkg file

Select file ⓘPrivileges.dmg

Name ⓘ

Privileges

Description ⓘ

SAP Privileges App

Publisher ⓘ

SAP

Category ⓘ

0 selected

Information URL ⓘ

Enter a valid url

Privacy URL ⓘ

Enter a valid url

Developer ⓘ

Owner ⓘ

Notes ⓘ

Logo ⓘ

Select image

Ignore app version ⓘ

Yes

No

Configure the app bundle identifiers and version numbers to be used to detect the presence of the app.

Included apps

i Provide the list of apps included in the uploaded file. The app list is case-sensitive. The app listed first is used as the primary app in app reporting. [Learn more about included apps.](#)

App bundle ID (CFBundleIdentifier)	App version (CFBundleShortVersionString)	
corp.sap.privileges	1.5.3	
Enter bundle ID	Enter app version	

Under **Assignments** click **+Add all devices**

VPP

Deploy Slack

In **Apple Business Manager**

In the sidebar click **Apps and Books**

Search for **Slack for Desktop** and select the macOS version

Under Buy Licenses select the location corresponding to your Intune instance and specify 10 in the quantity then click **Get**

In the **Intune UI**

Apps -> macOS

Select **Slack for Desktop** from the list of available apps. If it is not listed try clicking **Refresh**

Under **Properties -> Assignments** click **Edit** and click **+Add all devices**

Mac Additional Setup

Set Mac Wallpaper (Optional)

In order to use a custom image as the Mac Wallpaper the image file needs to be installed on the device. There are multiple methods to then set this image as the Wallpaper. This example uses a custom package to install the images and run a post install script to set the Wallpaper.

Create the .pkg file using a tool such as [Packages from Whitebox](#). Include a post install script with the following content:

```
#!/bin/zsh
osascript -e 'tell application "Finder" to set desktop picture to POSIX file "/usr/loca
```

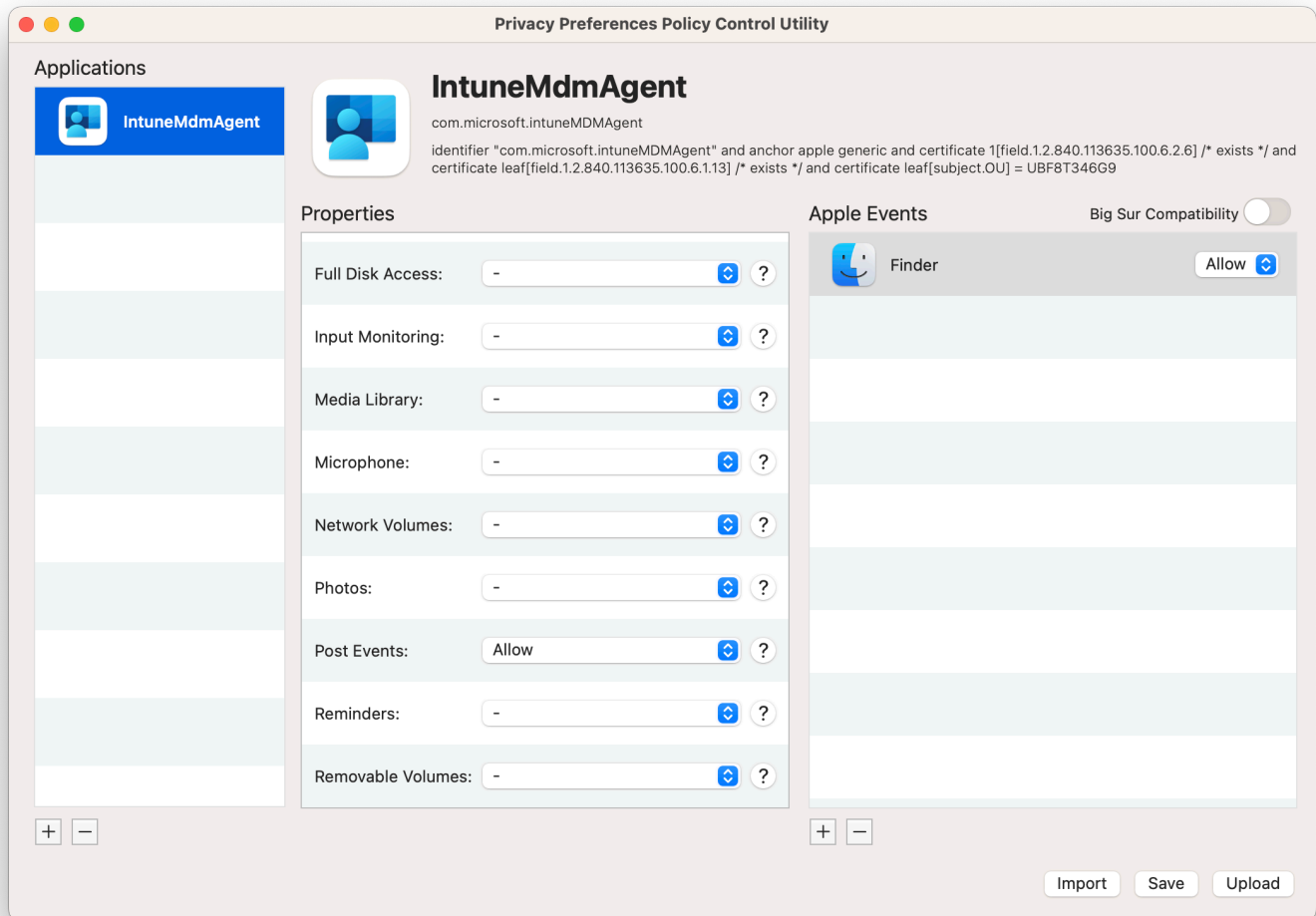
In the **Intune UI**

Apps -> macOS

Click **+ Add**

From the drop down menu select **Other -> macOS App (PKG)**

*The user would normally be prompted to approve the MDM Agent setting the Wallpaper so we deploy a PPPC Profile. This can be created using the [Jamf PPPC Utility](#) as follows. Save the profile as IntunePPPC.mobileconfig



In the **Intune UI**

Navigate to **Devices -> macOS -> Configuration Profiles**

Click **+ Create** to Create a new profile

Click **Profile Type -> Templates** then **Custom** then **Create**

Give the profile a name e.g. Intune Agent PPPC then click **Next**

For **Custom configuration profile name** enter Intune to Finder PPC

Set **Deployment Channel** to **Device Channel**

Upload the Custom Configuration profile file. Use filename IntunePPPC.mobileconfig

Under **Assignments** click **+Add all devices**

Compliance Policies

Disk Encryption Policy

*Intune can implement Compliance Policies which are evaluated as true or false. This calculated state can be used by Azure AD Conditional Access policies to determine whether a user can access certain cloud based**resources from the device.**

In the Intune UI

Navigate to **Devices -> macOS -> Compliance policies**

Click **+ Create policy** to Create a new policy

Give the profile a name e.g. Basic Compliance then click **Next**

Set the following:

Device Health -> Require System Integrity Protection to Require

System Security -> Device Security -> Firewall to Enable

System Security -> Device Security -> Incoming Connections to Block

System Security -> Device Security -> Stealth Mode to Enable

Mac compliance policy

macOS

- ✓ Basics
- 2 Compliance settings**
- 3 Actions for noncompliance
- 4 Assignments
- 5 Review + create

Device Health

Require system integrity protection ⓘ

Require

Not configured

Device Properties

Operating System Version

Minimum OS version ⓘ

13.6 ✓

Maximum OS version ⓘ

Not configured

Minimum OS build version ⓘ

Not configured

Maximum OS build version ⓘ

Not configured

System Security

Password

Require a password to unlock devices. ⓘ

Require

Not configured

Simple passwords ⓘ

Block

Not configured

Minimum password length ⓘ

4

Password type ⓘ

Device default ▼

Number of non-alphanumeric characters in password ⓘ

Not configured ▼

Maximum minutes of inactivity before password is required ⓘ

Not configured ▼

Password expiration (days) ⓘ

41

Number of previous passwords to prevent reuse ⓘ

5

Encryption

Require encryption of data storage on device. ⓘ

Require

Not configured

Device Security

Firewall ⓘ

Enable

Not configured

Incoming connections ⓘ

Block

Not configured

Stealth Mode ⓘ

Enable

Not configured

Gatekeeper

Allow apps downloaded from these locations ⓘ

Mac App Store and identified developers



Set the **Action** to **Mark device noncompliant** and the **Schedule** to **Immediately**

Specify the sequence of actions on noncompliant devices

Action	Schedule (days after noncompliance) ⓘ	Message template	Additional recipients (...)
Mark device noncompliant	Immediately		
	0		

Under Assignments click **+Add all devices**

Mac Device Assignment

In **Apple Business Manager**

From the sidebar click **Devices**

Search for the serial number of the Mac you will enrol and select it

Click **Edit MDM Server**

Click **Assign the Following MDM** and choose the appropriate MDM from the drop down list then click **Continue**

In the **Intune UI**

Navigate to **Devices -> macOS -> macOS Enrollment**

Click **Enrollment program tokens**

Select the token you created from the list

Under Manage click **Devices**

If the device you assigned from ABM is not listed click **Sync**, wait a minute and click **Refresh**

Check the box next to the device(s) in the list that you wish to apply an ADE profile then click **Assign Profile**

Under Assign Profile select the appropriate enrolment profile from the drop down list

Click **Sync** to sync with ABM. This can be done only after 15 mins or more from the last manual sync.

Enrolment Test

You can now test the enrolment process on the Mac you have assigned

Check profiles in the settings app.

Once Company Portal is installed launch it and log in. The device should appear non compliant.

Log out and back in again enabling FileVault in the process.

Log back into Company Portal and see the device is now compliant

Launch Apps, open <https://portal.office.com> in Safari and add Exchange account to Mail.

Mac Operations

Software Updates

In the **Intune UI**

Navigate to **Devices -> macOS -> Update policies for macOS**

Click **+ Create** to Create a new profile

Give the profile a name then click **Next**

Update policy behavior settings

Select how downloads, installations, and/or notifications should occur for each type of update.

Critical updates	Not configured
Firmware updates	Not configured
Configuration file updates	Not configured
All other updates (OS, built-in apps)	Install immediately

Update policy schedule settings

By default, when an update policy is assigned to a device, Intune deploys the latest updates at device check-in. You can instead create a weekly schedule with customized start and end times. If you choose to update outside of the scheduled time, Intune won't deploy updates until the scheduled time ends.

Schedule type ⓘ	Update at next check-in
-----------------	------------------------------------

DDM Software Updates

In the Intune UI

Navigate to Devices -> macOS -> Configuration Profiles

Click **+ Create** to Create a new profile

Click **Profile Type - Settings Catalogue** then **Create**

Give the profile a name e.g. macOS Software Update then click **Next**

Click **+ Add settings** and choose **Declarative Device Management (DDM) -> Software Update** then check Local Date and Time and Target OS Version and close the Settings Picker

These settings configure the declarations used by Apple's declarative device management feature. These settings are separate from older MDM settings and only apply to a device enabled for declarative management. [Learn more about declarative management at developer.apple.com](#)

Software Update

[Remove subcategory](#)

 2 of 4 settings in this subcategory are not configured

Target Local Date Time * ⓘ

31/01/2024



11:59 pm

Target OS Version ⓘ

14.3

FileVault Key Escrow

In the **Intune UI**

Navigate to **Devices -> macOS**

Select the relevant Mac from the list of devices

Click **Recovery Keys** then **Show Recovery Key**

Erase All Content and Settings

In the **Intune UI**

Navigate to **Devices -> macOS**

Select the relevant Mac from the list of devices

Click **X Erase**

Enter any Recovery PIN then click **Erase**

Additional Mac Config Profiles

When you run the Mac Evaluation Utility on a Mac there is a category Mac Management -> Security with a number of items that are evaluated. The following profiles can help achieve a Success status for some of these items.

Disable the Guest Account

Navigate to **Devices -> macOS -> Configuration Profiles**

Click **+ Create** to Create a new profile

Click **Profile Type -> Settings Catalogue** then **Create**

Give the profile a name e.g. Disable Guest Account then click **Next**

Click **+ Add settings** and choose **Accounts -> Accounts -> Disable Guest Account**

Then set **Disable Guest Account** to **True**

 Accounts [Remove category](#)

Accounts [Remove subcategory](#)

Configure the Accounts payload to enable or disable guest accounts.

 1 of 2 settings in this subcategory are not configured

Disable Guest Account   True 

Disable Auto-login

In the **Intune UI**

Navigate to **Devices -> macOS -> Configuration Profiles**

Click **+ Create** to Create a new profile

Click **Profile Type -> Templates** then **Custom** then **Create**

Give the profile a name e.g. Disable Auto Login then click **Next**

For **Custom configuration profile name** enter **Disable Auto Login**

Set **Deployment Channel** to **Device Channel**

Upload the Custom Configuration profile file

Assign to All Devices

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
  <key>PayloadContent</key>
  <array>
    <dict>
      <true/>
      <key>com.apple.login.mcx.DisableAutoLoginClient</key>
    </dict>
  </array>
  <key>PayloadDisplayName</key>
  <string>Login Window</string>
  <key>PayloadIdentifier</key>
  <string>com.example.myprofile</string>
  <key>PayloadType</key>
  <string>Configuration</string>
  <key>PayloadUUID</key>
  <string>61bd7d63-4a4a-4b67-9112-5ceb16afb4dc</string>
  <key>PayloadVersion</key>
  <integer>1</integer>
</dict>
</plist>
```

Configure Screensaver Lock

Navigate to **Devices -> macOS -> Configuration Profiles**

Click **+** **Create** to Create a new profile

Click **Profile Type -> Settings Catalogue** then **Create**

Give the profile a name e.g. Screensaver Lock then click **Next**

Click **+** **Add settings** and choose **System Configuration -> Screensaver -> Select all these settings**

Module Name =

Login Window Module Path = /System/Library/

Login Window Idle Time = *Preferred Value*

Ask For Password Delay = *Preferred Value*

Ask For Password = True

Configure Passcode Profile

NOTE: Be sure to set this via the Settings Catalogue and not via a Template

Navigate to **Devices -> macOS -> Configuration Profiles**

Click **+** **Create** to Create a new profile

Click **Profile Type -> Settings Catalogue** then **Create**

Give the profile a name e.g. Passcode Policy then click **Next**

Click **+** **Add settings** and choose **Security -> Passcode -> Needed Settings based on company security policy**

Configuration settings

^ Security

Passcode

Min Length ⓘ	8
Min Complex Characters ⓘ	2
Force PIN ⓘ	True

Resources

Setup

APNS

[Get Apple MDM push certificate](#)

Automated Device Enrolment

[Set up macOS enrollment](#)

Configuration Profiles

[Microsoft Enterprise SSO plug-in for Apple devices](#)

[macOS Configuration - See Settings for macOS](#)

[Jamf PPPC Utility](#)

[iMazing Profile Editor](#)

App Deployment

Apps and Books

[Volume-purchased iOS apps](#)

Scripting

[Use shell scripts on macOS devices in Microsoft Intune](#)

[Deploying macOS apps with the Microsoft Intune scripting agent](#)

[Sample shell scripts for macOS](#)

[Installomator](#)

Packaged Apps

[macOS LOB apps](#)

[AutoPkg](#)

[Munki - Software Management for OS X](#)

[Comprehensive guide to managing macOS with Intune](#) - A bit out of date but useful

[Packages App](#)

[Intune Uploader](#)

Operations

Software Update

[macOS update policies](#)

Third Party Tools

[SAP Privileges App](#)

[macOS Support App](#)

[Outset](#)